Math 445/545, Midterm II.

REMINDER: For a correct answer with no explanation you can get 0 points.

0. Write your name here:

1. Let $I = \langle 2+i \rangle$ be the ideal of the Gaussian integers $R = \mathbb{Z}[i]$ generated by $2+i$. What is characteristic of R/I?

Solution: We have $5 \cdot 1 = (2+i)(2-i) \in I$. Thus $5 \cdot 1 = 0$ in R/I. Thus the characteristic of R/I divides 5. We have $I \neq R$ since $2+i$ is not invertible in R. Thus $R/I \neq 0$ and the characteristic of R/I equals 5.

Answer: the characteristic of R/I equals 5.

Comment. In fact, $R/I \simeq \mathbb{Z}_5$. Show that!

2. Is the ideal $N = \langle x^2+2x+1 \rangle \subset \mathbb{Q}[x]$ maximal? How many ideals are strictly in between of N and $\mathbb{Q}[x]$?

Solution: the maximal ideals of $\mathbb{Q}[x]$ are of the form $\langle p \rangle$ where p is an irreducible polynomial; since $x^2+2x+1 = (x+1)^2$ is not irreducible the ideal $N = \langle (x+1)^2 \rangle$ is not maximal. If $N = \langle (x+1)^2 \rangle \subset M = \langle m \rangle$ then $(x+1)^2 \in N$ should be divisible by m; thus we have just 3 options for m: $1, x+1, (x+1)^2$ (all factors of $(x+1)^2$ up to units). Clearly $N = \langle (x+1)^2 \rangle$ and $\mathbb{Q}[x] = \langle 1 \rangle$; thus we have a unique ideal strictly in between of N and $\mathbb{Q}[x]$: $\langle 1+x \rangle$.

Answer: there is only one ideal strictly in between of N and $\mathbb{Q}[x]$: $\langle x+1 \rangle$.

3. Prove that $\mathbb{Z}_3[x]/\langle x^3+x^2-1\rangle$ is a field. How many elements are in this field?

Solution: $\mathbb{Z}_3$ is a field; so $\mathbb{Z}_3[x]/\langle p(x)\rangle$ is a field if and only if $p(x)$ is irreducible. The polynomial $p(x)=x^3+x^2-1\in\mathbb{Z}_3[x]$ is irreducible: it is of degree 3, so if it is not irreducible it would have a zero in $\mathbb{Z}_3$; it does not since $p(0)=-1=2, p(1)=1, p(2)=2$.

For any $p(x)$, $\mathbb{Z}_3[x]/\langle p(x)\rangle$ is a vector space over $\mathbb{Z}_3$ of dimension equal to the degree of $p(x)$. Thus $\mathbb{Z}_3[x]/\langle x^3+x^2-1\rangle$ has dimension 3, so it is isomorphic to $(\mathbb{Z}_3)^3$ as a vector space over $\mathbb{Z}_3$. In particular it has $3^3=27$ elements.

Answer: $\mathbb{Z}_3[x]/\langle x^3+x^2-1\rangle$ is a field with 27 elements.

4. Show that the rings $\mathbb{Z}[\sqrt{-2}]$ and $\mathbb{Z}[\sqrt{-3}]$ are not isomorphic.

Solution: Assume that $\phi:\mathbb{Z}[\sqrt{-2}]\to\mathbb{Z}[\sqrt{-3}]$ is an isomorphism. Then $x=\phi(\sqrt{-2})$ satisfies $x^2=\phi(\sqrt{-2})^2=\phi((\sqrt{-2})^2)=\phi(-2)=-2$. However for $x=a+b\sqrt{-3}$, $a,b\in\mathbb{Z}$ we have

$$x^2=(a+b\sqrt{-3})^2=a^2-3b^2+2ab\sqrt{-3},$$

and the equation $x^2=-2$ is equivalent to system of equations

$$\begin{cases} a^2-3b^2 & =-2 \\ 2ab & =0 \end{cases}$$

which clearly has no integer solutions.

5. True or false questions (2 points each). Give a complete explanation:
(a) element $8+i \in \mathbb{Z}[i]$ is irreducible.

Solution: This is FALSE since $8+i=(2-i)(3+2i)$. Less mysteriously: $8+i$ divides $(8+i)(8-i)=65$ but it does not divide neither 5 nor 13. Thus it is not prime, hence not irreducible since $\mathbb{Z}[i]$ is PID and each irreducible in PID is prime.

(b) polynomial $x^5+x+2 \in \mathbb{Z}[x]$ is irreducible.

Solution: This is FALSE since this polynomial has root -1, hence it is divisible by $x+1$.

(c) ring $\mathbb{Z}[x,y]$ is UFD.

Solution: This is TRUE. The ring $\mathbb{Z}$ is PID hence UFD. Also we have a theorem saying that if R is UFD then so is $R[x]$. Applying this to $R=\mathbb{Z}$ we get that $\mathbb{Z}[x]$ is UFD; applying the same theorem to $R=\mathbb{Z}[x]$ we get that $\mathbb{Z}[x,y]=R[y]$ is UFD.

(d) ring $\mathbb{Z}_{125}$ is PID.

Solution: This is FALSE since $\mathbb{Z}_{125}$ is not an integral domain; for instance $5 \cdot 25=0$.

(e) every vector space has a finite basis.

Solution: This is FALSE since some vector spaces are infinite dimensional, e.g. space of polynomials $F[x]$ over a field F.